

AERO40007 –Mechanics

Imperial College of Science, Technology & Medicine

Department of Aeronautics

2.5 The 200-N force is to be resolved into components along lines $a-a'$ and $b-b'$. (a) Determine the angle α using trigonometry knowing that the component along $a-a'$ is to be 150 N. (b) What is the corresponding value of the component along $b-b'$?

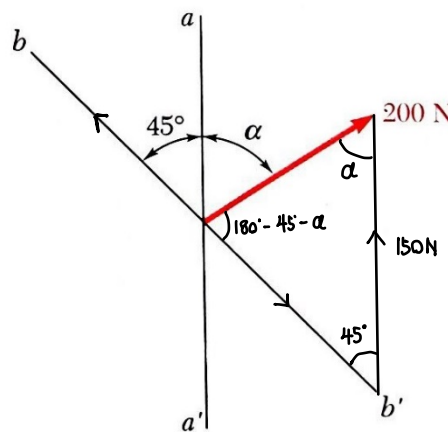


Fig. P2.5 and P2.6

$$(a) \quad \frac{150\text{N}}{\sin(135^\circ - \alpha)} = \frac{200\text{N}}{\sin 45^\circ}$$

$$\therefore \alpha = 102.97^\circ = 103^\circ$$

$$(b) \quad \frac{200\text{N}}{\sin 45^\circ} = \frac{F_{b-b'}}{\sin \alpha}$$

$$\therefore F_{b-b'} = 275.62\text{N} = 276\text{N}$$

2.38 Knowing that $\alpha = 50^\circ$, determine the resultant of the three forces shown.

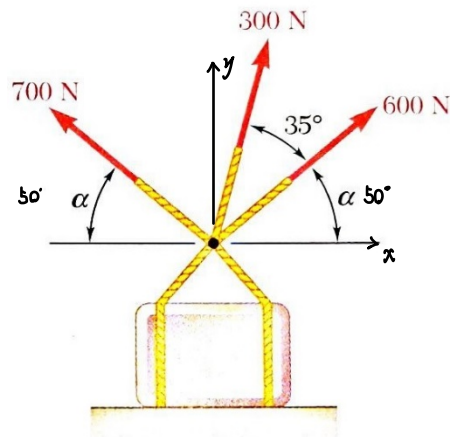


Fig. P2.38

$$\begin{aligned}
 F_x &= 600 \text{ N} \times \cos 50^\circ \\
 &\quad + 300 \text{ N} \times \cos 85^\circ \\
 &\quad - 700 \text{ N} \times \cos 50^\circ \\
 &= -38.13 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 F_y &= 600 \text{ N} \times \sin 50^\circ \\
 &\quad + 300 \text{ N} \times \sin 85^\circ \\
 &\quad + 700 \text{ N} \times \sin 50^\circ \\
 &= 1294.72 \text{ N}
 \end{aligned}$$

$$\therefore F = \sqrt{F_x^2 + F_y^2} = 1295.28 \text{ N} = 1295 \text{ N}$$

$$F_{\text{from } x} = 90^\circ + \tan^{-1} \left(\frac{F_x}{F_y} \right) = 91.69^\circ = 92^\circ$$

2.60 Two cables tied together at C are loaded as shown. Knowing that the maximum allowable tension in each cable is 900 N, determine (a) the magnitude of the largest force $\mathbf{P}$ which may be applied at C , (b) the corresponding value of α .

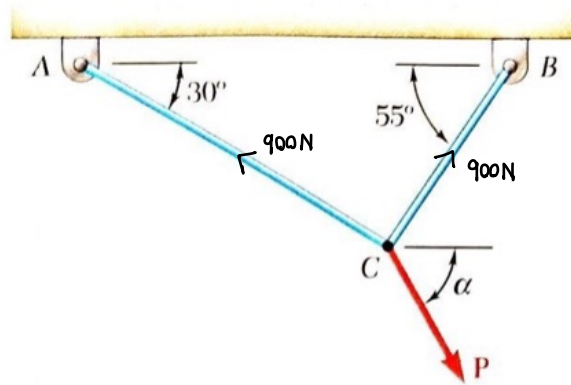


Fig. P2.60 and P2.61

$$\begin{aligned} (a) \quad P_x &= 900 \text{ N} \times \cos 30^\circ - 900 \text{ N} \times \cos 55^\circ \\ &= 263.20 \text{ N} \end{aligned}$$

$$\begin{aligned} P_y &= 900 \text{ N} \times \sin 30^\circ + 900 \text{ N} \times \sin 55^\circ \\ &= 1187.24 \text{ N} \end{aligned}$$

$$\therefore P = \sqrt{P_x^2 + P_y^2} = 1216.06 \text{ N} = 1216 \text{ N}$$

$$(b) \quad \alpha = \tan^{-1} \left(\frac{P_y}{P_x} \right) = 77.50^\circ = 78^\circ$$